Assignment 02 solution:

1a) $x_0 = -10 \quad x_1 = 20$

$y_0 = 5 \quad y_1 = 50$

$P(t) = (-10, 5) + t(30, 45)$

When $t = 3/4$,

$P(3/4) = (-10, 5) + \frac{3}{4}(30, 45)$

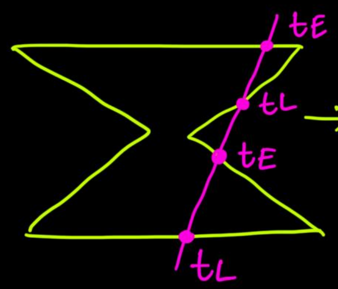
$= (12.5, 38.75)$

$P(7) = (-10, 5) + 7(30, 45)$

$= (200, 320)$ (Outside line segment)

b) Any concave shape:

e.g.



→ Explain in short why cyrus-beck will fail here.

c) $x_{min} = -10 \quad x_{max} = 50 \quad y_{min} = 10 \quad y_{max} = 150$

$x_0 = 30 \quad x_1 = 100 \quad y_0 = 40 \quad y_1 = 90$

$D = (70, 50) \quad t_E = 0 \quad t_L = 1$

Boundary	Ni	Ni·D	PE/PL	t	tE	tL
Left	(-1,0)	-70	PE	$\frac{-(30+10)}{70} = -0.57$	0	1
Right	(1,0)	70	PL	$\frac{-(30-50)}{70} = 0.29$	0	0.29
Top	(0,1)	50	PL	$\frac{-(40-150)}{50} = 2.2$	0	0.29
Bottom	(0,-1)	-50	PE	$\frac{-(40-10)}{50} = -0.6$	0	0.29

Since $t_E < t_L$, line needs to be clipped

2a) $x_{min} = -50 \quad x_{max} = 10 \quad y_{min} = -10 \quad y_{max} = 10$

$x_0 = -20 \quad x_1 = 5 \quad y_0 = -30 \quad y_1 = 20$

$O1 = 0100 \qquad O1 \text{ OR } O2 \;!= 0000 \qquad m = \frac{20+30}{5+20} = 2$

$O2 = 1000 \qquad O1 \text{ AND } O2 == 0000$

With O1:

Bottom $y = y_{min} = -10$

$x = -20 + \frac{1}{2}(-10+30)$

$= -10$

New $O1 \to 0000$

New O1 OR O2 != 0000

New O1 AND O2 == 0000

With O2:

Top $y = y_{max} = 10$

$x = 5 + \frac{1}{2}(10-20)$

$= 0$

$\therefore$ New end points $(-10, -10)$ to $(0, 10)$

b) Two scenarios: i) Very small region → Explain (trivial accept)

ii) Very large region → Explain (trivial reject)

3a) $\begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} \cos 30 & -\sin 30 & 0 \\ \sin 30 & \cos 30 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & -5 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1/6 & 0 & 0 \\ 0 & 1/6 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & -20 \\ 0 & 1 & -10 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & -8 & 0 \\ -7 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 12 & 8 & 11 \\ 15 & 10 & 17 \\ 1 & 1 & 1 \end{bmatrix}$

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3 vertices

(Multiplication from left to right)

$= \begin{bmatrix} 0.725 & -1.237 & 2.260 \\ -0.927 & -0.520 & -1.585 \end{bmatrix} \times \begin{bmatrix} 12 & 8 & 11 \\ 15 & 10 & 17 \end{bmatrix}$

$x = -20 + \frac{1}{2}(-10+30)$

$= -10$

New $01 \rightarrow 0000$

With 02:

Top $y = y_{max} = 10$

$x = 5 + \frac{1}{2}(10-20)$

$= 0$

$\therefore$ New end points $(-10,-10)$ to $(0,10)$

b) Two scenarios: i) Very small region $\rightarrow$ Explain (trivial accept)

ii) Very large region $\rightarrow$ Explain (trivial reject)

3a) $$\begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} \cos 30 & -\sin 30 & 0 \\ \sin 30 & \cos 30 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & -5 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1/6 & 0 & 0 \\ 0 & 1/6 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & -20 \\ 0 & 1 & -10 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & -8 & 0 \\ -7 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 12 & 8 & 11 \\ 15 & 10 & 17 \\ 1 & 1 & 1 \end{bmatrix}$$

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3 vertices

(Multiplication from left to right)

$$= \begin{bmatrix} 0.725 & -1.237 & 2.260 \\ -0.927 & -0.520 & -1.585 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 12 & 8 & 11 \\ 15 & 10 & 17 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -7.595 & -4.31 & -10.794 \\ -20.509 & -14.201 & -20.622 \\ 1 & 1 & 1 \end{bmatrix}$$ Original vertices

b) Collectively all are affine transformations, hence parallel lines are preserved.